

Install Jupyter Notebook in VS Code and Test OpenRouter

Step-by-step setup guide

This guide shows how to install Jupyter Notebook inside VS Code, create a clean Python environment, and test an OpenRouter API connection from a notebook.

1. Install Python

First check whether Python is already installed. Open Command Prompt or Terminal and run:

```
python --version
```

If that does not work, try:

```
python3 --version
```

If it shows something like Python 3.11.x or Python 3.12.x, continue. If not, install Python from the official Python website. On Windows, select Add Python to PATH during installation.

2. Install VS Code extensions

Open VS Code, go to Extensions, or press Ctrl + Shift + X. Install these extensions:

- Python
- Jupyter

3. Create a project folder

Create a folder anywhere, for example:

```
openrouter-jupyter-test
```

Open it in VS Code using File > Open Folder.

4. Open the VS Code terminal

Inside VS Code, open Terminal > New Terminal. Check Python again:

```
python --version
```

If needed, try:

```
python3 --version
```

5. Create a virtual environment

Windows

```
python -m venv .venv  
.venv\Scripts\activate
```

macOS or Linux

```
python3 -m venv .venv  
source .venv/bin/activate
```

After activation, your terminal should show `(.venv)`. That means the environment is active.

6. Install Jupyter and required packages

Run this inside the activated virtual environment:

```
pip install notebook jupyter ipykernel
```

Then install packages for OpenRouter and basic notebook work:

```
pip install openai python-dotenv pandas numpy matplotlib
```

7. Add the virtual environment as a notebook kernel

Run:

```
python -m ipykernel install --user --name openrouter-env --display-name "Python (OpenRouter Env)"
```

This makes your environment visible inside VS Code notebooks.

8. Create a new Jupyter Notebook

In VS Code, choose File > New File, then select Jupyter Notebook. Or create a file named:

```
test_openrouter.ipynb
```

9. Select the correct kernel

At the top-right of the notebook, click Select Kernel and choose:

```
Python (OpenRouter Env)
```

If you do not see it, restart VS Code.

10. Test basic Python in the notebook

In the first notebook cell, run:

```
print("Jupyter is working inside VS Code")
```

Run the cell with Shift + Enter. If you see output, Jupyter is working.

11. Create a .env file for OpenRouter

Inside your VS Code project folder, create a file named .env and add:

```
OPENROUTER_API_KEY=your_openrouter_api_key_here
```

Replace your_openrouter_api_key_here with your actual OpenRouter key. Do not paste your API key directly inside the notebook.

12. Test OpenRouter from the notebook

Create a notebook cell with this code:

```
import os
from dotenv import load_dotenv
from openai import OpenAI

load_dotenv()

client = OpenAI(
    base_url="https://openrouter.ai/api/v1",
    api_key=os.getenv("OPENROUTER_API_KEY")
)

response = client.chat.completions.create(
    model="openai/gpt-4o-mini",
    messages=[
        {"role": "user", "content": "Explain machine learning in 3 simple lines."}
    ]
)

print(response.choices[0].message.content)
```

Run it with Shift + Enter. If everything is correct, you should get a model response.

Recommended notebook structure

1. Setup and imports
2. Load API key

3. Create OpenRouter client
4. Send first prompt
5. Try different models
6. Build reusable function

Example reusable function:

```
def ask_openrouter(prompt, model="openai/gpt-4o-mini"):
    response = client.chat.completions.create(
        model=model,
        messages=[
            {"role": "user", "content": prompt}
        ]
    )
    return response.choices[0].message.content

answer = ask_openrouter("What is Agentic AI?")
print(answer)
```

Common errors and fixes

python is not recognized

Python is not added to PATH. Reinstall Python and select Add Python to PATH. On Windows, you can also try:

```
py --version
```

pip is not recognized

Use Python to call pip directly:

```
python -m pip install notebook jupyter ipykernel
```

VS Code notebook says Select Kernel

Click Select Kernel > Python Environments > Python (OpenRouter Env).

ModuleNotFoundError: No module named openai

Your notebook is likely using the wrong kernel. Select Python (OpenRouter Env), then install again if needed:

```
pip install openai
```

OpenRouter authentication error

Check your .env file:

```
OPENROUTER_API_KEY=sk-or-xxxxxxx
```

Then restart the notebook kernel.

Final command summary

Create environment

```
python -m venv .venv
```

Activate on Windows

```
.venv\Scripts\activate
```

Activate on macOS or Linux

```
source .venv/bin/activate
```

Install packages

```
pip install notebook jupyter ipykernel openai python-dotenv pandas numpy matplotlib
```

Register kernel

```
python -m ipykernel install --user --name openrouter-env --display-name "Python (OpenRouter Env)"
```

Then create test_openrouter.ipynb, select Python (OpenRouter Env), and run your OpenRouter code.